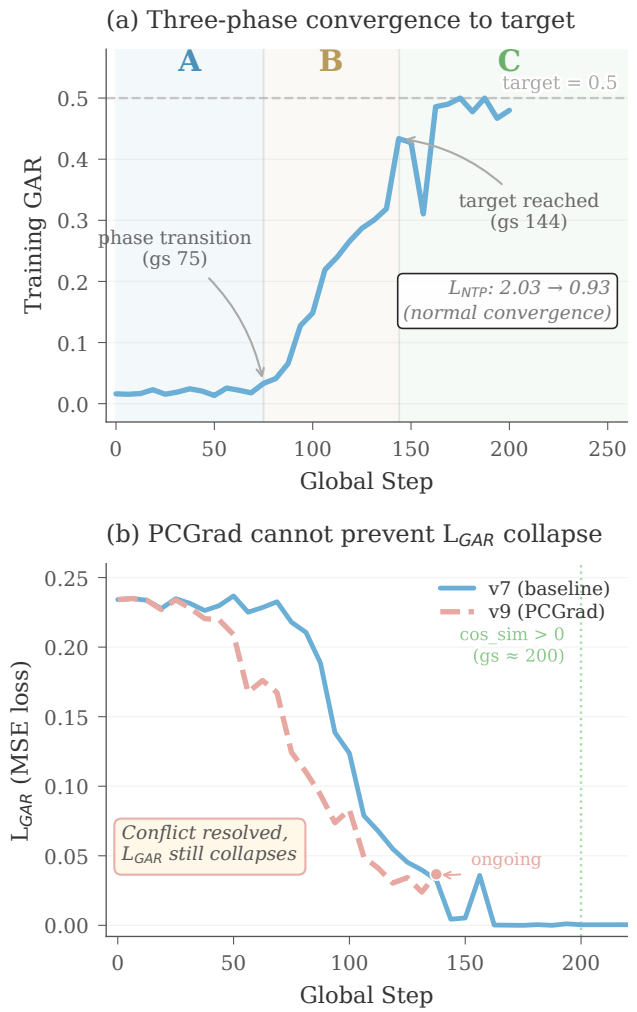
(a) Three-phase convergence to target
A
B
C
target = 0.5
Training GAR
0.0
0.1
0.2
0.3
0.4
0.5
phase transition
(gs 75)
target reached
(gs 144)
L_{NTP}: 2.03 → 0.93
(normal convergence)
0
50
100
150
200
250
Global Step
(b) PCGrad cannot prevent L_{GAR} collapse
v7 (baseline)
v9 (PCGrad)
cos_sim > 0
(gs ≈ 200)
L_{GAR} (MSE loss)
0.00
0.05
0.10
0.15
0.20
0.25
Conflict resolved,
L_{GAR} still collapses
ongoing
0
50
100
150
200
Global Step